

Rational points on modular curves

(Advancement to Candidacy)

Chris Xu
chx007@ucsd.edu

UC San Diego

April 7th, 2025

My research: equation solving

Given polynomial $f(x, y) = 0$, find its rational solutions!

That is, given a smooth projective curve $X/\mathbf{Q}$, find $X(\mathbf{Q})$.

Example

Let $f(x, y) := x^2 + y^2 - 1$.

The rational solutions classify Pythagorean triples.

Example

Let $f_n(x, y) := y^2 - x^3 + n^2x$ for $n \in \mathbf{N}$.

The rational solutions classify right triangles of area n with rational side lengths.

Example

Let $f(x, y) := y^4 + 5x^4 - 6x^2y^2 + 6x^3 + 26x^2y + 10xy^2 - 10y^3 - 32x^2 - 40xy + 24y^2 + 32x - 16y$. The rational solutions classify elliptic curves with “non-split level 13 structure”.

Trichotomy by genus

Around the 1920s, it emerged that curves could be put into three classes of increasing complexity: genus 0, genus 1, and genus ≥ 2 .

My research deals with explicit methods for determining rational points in the genus ≥ 2 case.

Genus 0

In this case, everything is $\mathbf{P}^1$ when viewed over $\mathbf{C}$. If a rational point exists, all other rational points are determined by rational slope chords.

Example

Consider $x^2 + y^2 = 1$.

Take $(1, 0)$. Then all other points are determined by rational slope lines emanating from $(1, 0)$.

Genus 1

Elliptic curves E . There is a group law (“three collinear points sum to zero”).

Theorem (Mordell, 1922)

We have $E(\mathbf{Q}) \cong \mathbf{Z}^r \oplus T$ for some r , where T is finite.

The group T has been completely classified (Mazur). There is an algorithm to compute $E(\mathbf{Q})$. But r remains very mysterious.

Example

Consider $y^2 = x^3 - n^2x$ as n varies over integers. Note that they are all isomorphic to each other over $\mathbf{C}$, but not over $\mathbf{Q}$! (More on this later.) Recent progress on the behavior of r by A. Smith (2017-2022).

Genus ≥ 2

Curves “of general type”.

Theorem (Faltings, 1983)

Suppose $X/\mathbf{Q}$ has genus at least 2. Then $X(\mathbf{Q})$ is finite.

The most advanced tool we have for *determining* the rational solutions is a family of methods called **(depth n) Chabauty**, for $n \geq 1$.

Example

Consider $X: y^4 + 5x^4 - 6x^2y^2 + 6x^3 + 26x^2y + 10xy^2 - 10y^3 - 32x^2 - 40xy + 24y^2 + 32x - 16y = 0$, of genus 3. Determination of $X(\mathbf{Q})$ only happened in 2019 using quadratic Chabauty ($n = “1 + \varepsilon”$).

Chabauty's original theorem

Choose $b \in X(\mathbf{Q})$ and a prime p . Let J be the Jacobian¹ of X . There is a map $\text{AJ}_b: X \rightarrow J$ given by $\text{AJ}_b(x) := [x - b]$.

$$\begin{array}{ccc} X(\mathbf{Q}) & \hookrightarrow & X(\mathbf{Q}_p) \\ \downarrow \text{AJ}_b & & \downarrow \text{AJ}_b \\ J(\mathbf{Q}) & \hookrightarrow & J(\mathbf{Q}_p) \end{array}$$

Chabauty's idea: if $J(\mathbf{Q})$ is *not* Zariski dense in $J(\mathbf{Q}_p)$, then $J(\mathbf{Q}) \cap X(\mathbf{Q}_p)$ is a finite set containing $X(\mathbf{Q})$.

Theorem (Chabauty, 1941)

Let $g := \dim(J)$ and $r := \text{rk}_{\mathbf{Z}}(J(\mathbf{Q}))$.

Then $X(\mathbf{Q})$ is finite, provided that $r < g$ holds.

¹The Jacobian is an abelian variety parametrizing the degree 0 divisors on X . Its dimension is g , the genus of X . An abelian variety is a “higher dimensional elliptic curve”. There is a generalization of Mordell-Weil to abelian varieties.

Coleman's algorithmic interpretation

In the 1980s, Coleman makes Chabauty's ideas into an algorithm by introducing a gadget called a **Coleman integral**².

First, consider the diagram

$$\begin{array}{ccc}
 X(\mathbf{Q}) & \hookrightarrow & X(\mathbf{Q}_p) \\
 \downarrow \text{AJ}_b & & \downarrow \text{AJ}_b \\
 J(\mathbf{Q}) & \hookrightarrow & J(\mathbf{Q}_p) \xrightarrow{\log} H^0(X, \Omega^1)^\vee
 \end{array}
 \quad \text{where } \log(D) := \left[\omega \mapsto \int_D \omega \right]$$

Next, find a basis of *annihilating differentials*³ $\omega_1, \dots, \omega_{g-r}$.

Finally, $X(\mathbf{Q})$ is contained in the finite set given by the locus

$$\int_b^t \omega_1 = \dots = \int_b^t \omega_{g-r} = 0. \quad (\text{all are power series in the variable } t.)$$

²A Coleman integral $\int_D \omega$ pairs a divisor $D \in J(\mathbf{Q}_p)$ with a 1-form ω . It behaves like an antiderivative if D is all on a single mod p “residue disk”, and otherwise one “extends by Frobenius”.

³i.e. a 1-form ω such that $\int_D \omega = 0$ for any $D \in J(\mathbf{Q})$.

Kim's nonabelian refinement

What if we do not have $r < g$? In the 2000s, Kim proposes replacing J with a “depth n Selmer variety” $\text{Sel}(U_n)$ associated to U_n ⁴.

$$\begin{array}{ccc}
 X(\mathbf{Q}) & \hookrightarrow & X(\mathbf{Q}_p) \\
 \downarrow \text{AJ}_{b,n} & & \downarrow \text{AJ}_{b,n} \quad \swarrow \text{dotted} \\
 \text{Sel}(U_n) & \hookrightarrow & H_f^1(\text{Gal}_{\mathbf{Q}_p}, U_n) \xrightarrow{\text{D}_{\text{dR}}} U_n^{\text{dR}} / \text{Fil}^0
 \end{array}$$

Here, $\text{AJ}_{b,n}$ sends x to a “non-abelian” path $[P_{x,b}]$, which is a torsor of $U_n = P_{x,x}$. The condition $r < g$ gets weaker as n increases.

Conjecture (Kim, 2005)

For any curve $X/\mathbf{Q}$, there is a large enough n such that $\text{Sel}(U_n)(\mathbf{Q}_p) \cap X(\mathbf{Q}_p) = X(\mathbf{Q})$.

⁴Consider the $\mathbf{Q}_p$ -algebraic fundamental group of the Tannakian category of finite étale covers (resp. unipotent connections) on $X_{\mathbf{C}_p}$. Then U_n (resp. U_n^{dR}) is its maximal n -unipotent quotient.

Algorithmic interpretations of Chabauty-Kim

Consider the diagram:

$$\begin{array}{ccc}
 X(\mathbf{Q}) & \hookrightarrow & X(\mathbf{Q}_p) \\
 \downarrow \text{AJ}_{b,n} & & \downarrow \text{AJ}_{b,n} \\
 \text{Sel}(U_n) & \hookrightarrow & H_f^1(\text{Gal}_{\mathbf{Q}_p}, U_n) \xrightarrow{\text{D}^{\text{dR}}} U_n^{\text{dR}} / \text{Fil}^0
 \end{array}$$

Making Chabauty-Kim explicit relies on finding coordinates on the variety $U_n^{\text{dR}} / \text{Fil}^0$. You end up with n -fold Coleman integrals.

- (1985) For $n = 1$, the 1-forms give coordinates on $H^0(X, \Omega^1)^\vee$.
- (2016) The **quadratic Chabauty** method of Balakrishnan-Dogra gives a coordinate for $n = "1 + \varepsilon"$ coming from arithmetic intersection theory.
- (2021) Corwin's **motivic Chabauty** gives a coordinate for arbitrary n using a “universal cocycle evaluation map”.

Twists

Over $\mathbf{C}$, elliptic curves E are classified by their j -invariant, $j(E) \in \mathbf{C}$. However, elliptic curves can be isomorphic over $\mathbf{C}$ but not over $\mathbf{Q}$. This is the phenomenon of twists, caused by the fact that $A_E := \text{Aut}_{\mathbf{C}}(E)$ is nontrivial:

$$A_E = \begin{cases} \mathbf{Z}/6\mathbf{Z} & j(E) = 0 \\ \mathbf{Z}/4\mathbf{Z} & j(E) = 1728 \\ \mathbf{Z}/2\mathbf{Z} & j(E) \notin \{0, 1728\}. \end{cases}$$

Elliptic curves with complex multiplication

If we consider the endomorphism ring $\text{End}_{\mathbf{C}}(E)$, we find that it is either $\mathbf{Z}$ or an order $\mathcal{O}$ in an imaginary quadratic field K .

We say that E has CM in the latter case. (For example, E has CM when $j(E) \in \{0, 1728\}$.)

Theorem (Early 1900s)

The maximal abelian extension of K is obtained by first adjoining $j(E)$, and then the coordinates of all the torsion points of E .

Torsion points on elliptic curves

For an elliptic curve E over $\mathbf{C}$, it's a torus. So the group of N -torsion points $E[N](\mathbf{C})$ is just $(\mathbf{Z}/N\mathbf{Z})^2$.

Over $\mathbf{Q}$, there is now an action of $\text{Gal}_{\mathbf{Q}}$. So we get a map $\rho: \text{Gal}_{\mathbf{Q}} \rightarrow \text{GL}_2(\mathbf{Z}/N\mathbf{Z})$.

If ρ is non-CM then its image will generally be close to being surjective.

If ρ is CM then its image will be very close to being abelian (cf. previous slide).

Moduli spaces of elliptic curves

Let N be a positive integer and let H be a subgroup of $\mathrm{GL}_2(\mathbf{Z}/N\mathbf{Z})$. The modular curve Y_H parametrizes elliptic curves with H -level structure:

$$Y_H(\bar{k}) := \{(j(E), HgA_E) : j(E) \in \bar{k}, g \in \mathrm{GL}_2(\mathbf{Z}/N\mathbf{Z})\}$$

$$Y_H(k) := \{(j(E), HgA_E) : j(E) \in k, HgA_E = Hg\rho(\mathrm{Gal}_k)A_E\}.$$

The modular curve X_H is a compactification of Y_H , obtained by adding some finite number of “cusps”.

Remark

Given an element of $X_H(k)$, you can always twist it so that the image of Galois lies in H .

Mazur's Program B

We can vary across all N to get a map⁵ $\rho: \text{Gal}_{\mathbf{Q}} \rightarrow \text{GL}_2(\hat{\mathbf{Z}})$.

Theorem (Serre)

The image of ρ is open in $\text{GL}_2(\hat{\mathbf{Z}})$.

You are led naturally to the following problem: classify all possible images of ρ for non-CM elliptic curves E . This is Mazur's Program B, proposed in the 1970s:

Classify the non-CM, non-cuspidal rational points of X_H , for open subgroups⁶ $H \leq \text{GL}_2(\hat{\mathbf{Z}})$.

In what follows, let us outline our proposed approach to resolve Mazur's Program B once and for all.

⁵Here, $\hat{\mathbf{Z}} := \varprojlim_N \mathbf{Z}/N\mathbf{Z}$, where the transition maps are the usual surjections $\mathbf{Z}/M\mathbf{Z} \twoheadrightarrow \mathbf{Z}/N\mathbf{Z}$ for $N \mid M$.

⁶Any open subgroup H comes from $H' \leq \text{GL}_2(\mathbf{Z}/N\mathbf{Z})$ for some integer N .

Mazur's Program B: initial reduction

First, it suffices to consider the open subgroups $H \leq \mathrm{GL}_2(\hat{\mathbf{Z}})$ that are maximal with respect to the property “ X_H has genus at least 2”.

There are infinitely many primes p , so shouldn't there be infinitely many such H ?

We have run into our first snag.

Conjecture (Serre's uniformity question)

There is a constant C such that for all primes $p > C$ and all subgroups $H \leq \mathrm{GL}_2(\mathbf{F}_p)$, the set $X_H(\mathbf{Q})$ consists entirely of CM points and cusps.

If Serre uniformity holds, then there are only finitely many open subgroups $H \leq \mathrm{GL}_2(\hat{\mathbf{Z}})$ that are maximal with respect to the property “ X_H has genus at least 2, and $X_H(\mathbf{Q})$ consists of CM points and cusps”. (More on this in a bit.)

Progress on Serre's uniformity question

If $H \leq \mathrm{GL}_2(\mathbf{F}_p)$ is a maximal subgroup, then one of the following holds. Text in this color describes known necessary conditions for $X_H(\mathbf{Q})$ to contain an “exceptional” point.

- ① H contains $\mathrm{SL}_2(\mathbf{F}_p)$. X_H is not even defined over $\mathbf{Q}$.
- ② The image of H in $\mathrm{PGL}_2(\mathbf{F}_p)$ is S_4 , A_4 or A_5 . X_H must have $p \leq 13$ (Serre, 1972).
- ③ (“Borel”) $H = \left\{ \begin{bmatrix} * & * \\ 0 & * \end{bmatrix} \right\}$. $X_H =: X_0(p)$ must have $p \leq 163$ (Mazur, 1978).
- ④ (“Normalizer of split Cartan”) $H = \left\{ \begin{bmatrix} * & 0 \\ 0 & * \end{bmatrix} \right\} \cup \left\{ \begin{bmatrix} 0 & * \\ * & 0 \end{bmatrix} \right\}$.
 $X_H =: X_s^+(p)$ must have $p \leq 13$ (Bilu-Parent-Rebolledo, 2013).
- ⑤ (“Normalizer of nonsplit Cartan”) $H = \left\{ \begin{bmatrix} x & \varepsilon y \\ y & x \end{bmatrix} \right\} \cup \left\{ \begin{bmatrix} x & \varepsilon y \\ -y & -x \end{bmatrix} \right\}$
 where $\varepsilon \in \mathbf{F}_p^\times$ is any non-square (H does not depend on the choice of ε). $X_H =: X_{ns}^+(p)$. Nothing is known!!

Serre uniformity in the nonsplit Cartan case: an idea

In the split Cartan case, Bilu-Parent need the following ingredients.

Text in this color denotes how you might adapt it to the non-split Cartan case.

- Construct a nontrivial modular unit⁷, integral over $\mathbf{Z}[j]$. For $X_{ns}^+(p)$, only one $\text{Gal}_{\mathbf{Q}}$ -orbit of cusps. Instead you probably have to use CM points. Use Gross-Zagier to construct modular functions f_i supported at Heegner divisors.
- Guarantee that a putative rational point does not intersect the cuspidal divisor. Hope that you need only finitely many f_i such that for any putative rational point, there is some i such that it does not intersect the divisor $D(f_i)$.
- Lower bounds on $|j(E)|$ for E non-CM, coming from bounds on the smallest degree of an isogeny between two isogenous elliptic curves. Non-split case has a somewhat more difficult moduli interpretation cf. Rebolledo-Wuthrich 2017.

⁷poles and zeroes only at cusps

Maximal subgroups of $g \geq 2$ with exceptional points

Now suppose Serre uniformity holds in the affirmative.

Suppose $H \leq \mathrm{GL}_2(\hat{\mathbf{Z}})$ is maximal of genus ≥ 2 with level

$N = p_1^{e_1} \cdots p_m^{e_m}$. Induct on m to classify all possible H , as follows.

- ① Enumerate the finitely many H of genus ≤ 1 .
- ② If $m = 1$, say $N = p^e$, use

$$0 \rightarrow I(p) \rightarrow \mathrm{GL}_2(\mathbf{Z}_p) \rightarrow \mathrm{GL}_2(\mathbf{F}_p) \rightarrow 0$$

and the fact that $I(p)$ is pro- p to compute the (finitely many) maximal open subgroups of $\mathrm{GL}_2(\mathbf{Z}_p)$ with genus ≥ 2 whose reduction mod p is a subgroup of $\mathrm{GL}_2(\mathbf{F}_p)$ containing an “exceptional point”.

- ③ Suppose $m > 1$. For $1 \leq i \leq m$ let $N^{(i)} := N/p_i^{e_i}$. By assumption on the level N of H , the projections of H onto $\mathrm{GL}_2(\mathbf{Z}/N^{(i)}\mathbf{Z})$ all have genus ≤ 1 , and then a “generalized Goursat’s lemma” tells you what H can be.

See Zywinia’s open image papers for an alternate formulation.

Motivation for equationless Chabauty

We now have finitely many H for which we need to find $X_H(\mathbf{Q})$.

You could now proceed by computing projective models for each X_H and applying Chabauty.

However, you are left wondering if there is a more natural approach that uses the moduli interpretation of X_H , instead of having to manually account for the ramification.

The answer is yes! But you will have to do some precision analysis instead. This approach is called “equationless” or “model-free” Chabauty.

Expressing large Coleman integrals as tiny ones

The difficulty of linear Chabauty boils down to computing Coleman integrals.

If you had a plane curve, you first compute large Coleman integrals into tiny Coleman integrals using a Frobenius lift.

For modular curves, you can use the Hecke correspondence instead. This works because of the Eichler-Shimura relation.

You get the following formula of column vectors:

$$\left[\int_a^b \omega_i \right]_i^T = (p+1-A)^{-1} \left[\sum_{j=1}^{p+1} \left(\int_{b_j}^b \omega_i - \int_{a_j}^a \omega_i \right) \right]_i^T$$

where ω_i is a basis of annihilating differentials, A is the matrix of T_p acting on the ω_i , and $T_p([a]) =: [a_1 + \cdots + a_{p+1}]$ (crucially, note that the a_j all lie in the same disk as a).

Computing tiny Coleman integrals

This is easy in the plane curve case: find a uniformizer t on the residue disk, and then compute $\int_a^b \omega = \int_a^b f(t) dt$ by formally antidifferentiating each term of $f(t) \in \mathbf{Q}_p[[t]]$.

For modular curves, this is harder: you only have the j -invariant as a coordinate.

You can take a q -expansion of ω and then write it in terms of the uniformizer $j - j_0$ using analytic methods, but then you will have to pin down the coefficients as algebraic numbers.

But this is actually feasible! You can study the ramification of $j: X_H \rightarrow \mathbf{P}^1$ to figure out the denominators of the coefficients. Then you can use integer programming or Fourier-theoretic techniques to pin down the algebraic integers rigorously. (Rendell-X., 2025)

Remark

My paper requires you are expanding at a CM point, but this is OK because each residue disk has lots of CM points.

What is quadratic Chabauty?

Here is how quadratic Chabauty works. Let S be the primes of bad reduction for $X/\mathbf{Q}$. Fix a basepoint $b \in X(\mathbf{Q})$.

For $x \in X(\mathbf{Q})$, we have an identity $h(x) = h_p(x) + \sum_{v \in S} h_v(x)$, where h , h_p and h_v are certain functions

$$h: J(\mathbf{Q}) \rightarrow \mathbf{Q}_p, \quad h_p: X(\mathbf{Q}_p) \rightarrow \mathbf{Q}_p, \quad h_v: X(\mathbf{Q}_v) \rightarrow \mathbf{Q}_p.$$

The function h then p -adically interpolates to a function

$$\tilde{h}: J(\mathbf{Q}_p) \rightarrow \mathbf{Q}_p.$$

The computation of h_p yields some double Coleman integrals.

On the other hand, h_v lands in a computable finite set $\mathcal{T}$.

So one only needs to find the solutions to the $\#\mathcal{T}$ equations

$$\tilde{h} \circ \text{AJ}_b - h_p - \alpha = 0 \quad (\alpha \in \mathcal{T})$$

to get a finite set containing $X(\mathbf{Q})$.

Computation of h , the global height pairing

We have the equations

$$\tilde{h} \circ \text{AJ}_b - h_p - \alpha = 0 \quad (\alpha \in \mathcal{T}).$$

If you have enough rational points, you can compute $\tilde{h} \circ \text{AJ}_b$ simply by “interpolation from h_p ”.

You may not have enough rational points. But for modular curves, you have an out! The function h comes from a certain p -adic height pairing⁸, which can be computed directly via p -adic Gross-Zagier formulae.

The case $X_0^+(N)$ is done in (Hashimoto, 2022). We hope to generalize this to arbitrary modular curves.

⁸The functions h_p and h_v come from the local contributions to the height pairing.

Computation of h_v

This requires a semistable model of X at v .

For modular curves, the way to make this equationless is to use (Weinstein, 2016), which gives semistable models of arbitrary level modular curves.⁹

We hope to turn Weinstein's techniques into an algorithm.

⁹So long as $v \neq 2$. Sigh.

Computation of h_p (one aspect of it)

The most salient issue is the conversion of large double Coleman integrals into tiny Coleman integrals.

For modular curves, you cannot use the Hecke correspondence T_p anymore; you have to use an actual function. The map V_p (quotienting an elliptic curve by its canonical subgroup) provides a candidate, provided you pick CM points that are “not too supersingular at p ”.

You have a path γ from P to Q which you rewrite as $c_a \circ V_p(\gamma) \circ c_b$, where c_a goes from P to $V_p(P)$ and c_b goes from $V_p(Q)$ to Q .

For Hecke eigenforms ω_1, ω_2 , you have $V_p^* \omega_i = \alpha_i \omega_i + df_i$ for $i = 1, 2$.

You can then combine this with the *shuffle relation*

$$\int_{\gamma} \omega_1 \omega_2 = \int_{c_a} \omega_1 \omega_2 + \int_{c_b} \omega_1 \omega_2 + \int_{V_p(\gamma)} \omega_1 \omega_2 + \int_{c_a} \omega_1 \int_{V_p(\gamma)} \omega_2 + \int_{c_a} \omega_1 \int_{c_b} \omega_2 + \int_{V_p(\gamma)} \omega_1 \int_{c_b} \omega_2$$

to get an expression in terms of single integrals and tiny double integrals.

Some speculation on motivic Chabauty

Corwin's theory gives a coordinate for arbitrary depth n . He states:

“We hope to work with [Jacobians of arbitrary curves] in the future, the only obstacle being the messiness of the representation theory of reductive groups larger than GL_2 .”

In general, you have to work with GSp_{2g} .

But for a modular curve, all simple factors of its Jacobian are of GL_2 -type. For us, it means that you only have to work with $\mathrm{Res}_{K/\mathbf{Q}} \mathrm{GL}_2$ for certain number fields K .

We hope to make some progress on this matter.¹⁰

¹⁰Unfortunately, there is only so much you can do in a few years...

Thanks for listening!